

Master Framework: Connect / Relationship / Hardware Pipeline

Project Context

CabinetNC unified plugin is building a manufacturing-oriented cabinet workflow inside Fusion 360.

The Connect module must not be a single ad-hoc feature. It should become the foundation for:

```
panel relationship detection
→ relationship verification
→ hardware feature intent
→ Fusion cut execution
→ machining metadata writeback
```

Current state:

- Board Relationship bbox/AABB candidate detection exists.
- relationships.* routes exist.
- Developer Debug UI exists.
- Hardware screw-hole preview from BoardRelationship exists.
- Offline regression passes.
- Existing generator behavior is unchanged.
- Existing hardware cut behavior is unchanged.
- Real Fusion cut from relationship is not enabled.
- bbox_candidate relationships are safeForPreview=true but safeForCut=false.

Core Pipeline

All future development must follow this pipeline:

```
Panel Metadata
↓
PanelSnapshot[]
↓
RelationshipService
↓
BoardRelationship[]
↓
Verification / Confirmation
↓
VerifiedRelationship
↓
HardwareRuleEngine
```

```
↓  
HardwareFeatureIntent[]  
↓  
FusionExecution  
↓  
Operation Metadata / Panel Metadata
```

Do not skip layers.

Layer 1: Panel Metadata

Panel bodies are the input source.

Required identifiers:

```
panelId  
bodyName  
sourceBoardId  
boardType  
role  
materialClass  
bbox  
dimensions  
faceRegistry, when available
```

Do not rely on entityToken as the only stable identifier.

Layer 2: Relationship Detection

Current bbox/AABB detection is only a candidate detector.

It may classify:

```
edge_to_surface  
surface_to_surface  
gap_parallel  
intersection  
none
```

All bbox/AABB relationships must be marked:

```
{  
  "detectionMethod": "bbox_aabb",  
  "verification": {
```

```

"level": "bbox_candidate",
"safeForPreview": true,
"safeForCut": false,
"requiresManualConfirmation": true
}
}

```

Do not treat bbox_candidate as a physical truth.

Layer 3: Relationship Verification

Relationship verification levels are:

```

verification.level:
| "bbox_candidate"
| "manual_confirmed"
| "face_verified"
| "generator_declared"
| "cut_approved"

```

Meaning:

```

bbox_candidate:
    Detected by bbox/AABB only.
    Can be previewed.
    Cannot be cut.

manual_confirmed:
    User explicitly confirmed this relationship for cut testing.
    Can be cut.
    Not production-true by itself.

face_verified:
    Real BRep / face-level check has confirmed face contact.
    Can be cut if host/target and entry face are valid.

generator_declared:
    Generator declared this intended joint.
    Can be cut if geometry validation passes.

cut_approved:
    Final approved relationship for machining operation.

```

Cut route must reject any relationship where:

```
verification.safeForCut !== true
```

Layer 4: Hardware Rule Engine

Hardware must consume `BoardRelationship`.

Hardware must not independently rediscover board contact.

Allowed input:

```
BoardRelationship  
VerifiedRelationship
```

Output:

```
HardwareFeatureIntent {  
  schemaVersion: 1  
  featureId: string  
  type: "screw_hole" | "hinge_hole" | "lock_cutout" | "runner_hole" | "tongue" | "groove"  
  sourceRelationshipId: string  
  hostPanelId: string  
  targetPanelId?: string  
  geometry: object  
  source: {  
    method: "relationship_based_rule"  
    ruleId: string  
  }  
  validation: {  
    ok: boolean  
    warnings: string[]  
    errors: string[]  
  }  
}
```

Current supported hardware type:

```
screw_hole_from_edge_to_surface_v1
```

Do not implement hinge, lock, runner, tongue/groove until the screw-hole pipeline is stable.

Layer 5: Fusion Execution

Fusion execution must be separate from preview.

Preview route:

```
hardware.previewScrewHolesFromRelationship
```

Cut route:

```
hardware.createScrewHolesFromRelationship
```

Rules:

- Preview can accept `bbox_candidate`.
- Cut must require `safeForCut=true`.
- Cut must resolve host body by `hostPanelId`.
- Cut must only modify host body.
- Cut must not modify target body.
- Use `participantBodies` or equivalent Fusion API safety mechanism.
- Return JSON audit report.

Layer 6: Metadata Writeback

First write metadata to the Fusion cut feature.

Suggested metadata:

```
{
  "operationType": "SCREW_HOLE_FROM_RELATIONSHIP",
  "sourceRelationshipId": "...",
  "hostPanelId": "...",
  "targetPanelId": "...",
  "ruleId": "screw_hole_from_edge_to_surface_v1",
  "holeCount": 2,
  "diameterMm": 4,
  "depthMm": 15
}
```

Do not update body-level `features[]` unless a safe helper already exists.

Later milestone:

```
cut feature metadata
↓
panel metadata features[]
```

↓
millingSurfaceSvg / CNC intent

Developer UI Role

The current UI is a Developer Debug UI, not a production UI.

Allowed UI actions:

Create Relationship Fixture
Scan Relationships
Inspect Selected Pair
Preview Screw Holes
Confirm Relationship For Cut
Create Screw Holes From Confirmed Relationship
Show JSON Report
Clear Result

Do not build final product UI yet.

Do not build relationship graph, table editor, formal hardware editor, or complex workflow UI yet.

Development Milestones

Milestone 1: Candidate Layer

Already mostly complete.

- bbox/AABB relationship detection
- routes
- fixture
- offline regression
- debug UI
- hardware preview from relationship

Milestone 2: Verification Gate

Next priority.

- Add manual_confirmed level
- Add confirm relationship helper
- Add Developer UI confirm button

- Confirmed relationship sets safeForCut=true
- bbox_candidate remains blocked from cut

Milestone 3: Screw-Hole Cut Execution

After Milestone 2.

- Add hardware.createScrewHolesFromRelationship
- Require safeForCut=true
- Cut host body only
- Write cut feature metadata
- Return JSON audit
- Fusion smoke test

Milestone 4: Real Cabinet Smoke Test

Use real generator output.

- Generate Overhead or General Tall
- Scan relationships
- Preview screw holes
- Confirm one relationship manually
- Cut host body
- Verify target body is not modified
- Verify cut metadata

Milestone 5: Face-Level Verification

After screw-hole cut pipeline works.

- Use faceRegistry / BRep face data
- Detect face pairs
- Check normals
- Check plane distance
- Check overlap region
- Mark relationship as face_verified

Milestone 6: Generator-Declared Relationships

Later.

- Generators declare intended joints
- RelationshipService validates declared joints
- generator_declared + geometry check can become cut-safe

Milestone 7: Formal Connect UI

Only after backend pipeline is stable.

- Relationship list
- relationship confidence
- preview
- confirm
- cut
- metadata status

Do Not Do Yet

Do not implement these until the screw-hole connect pipeline is stable:

- hinge automation
- lock automation
- drawer runner automation
- tongue/groove automation
- full BRep verification for all cases
- formal customer-facing UI
- automatic production cut from bbox_candidate
- generator-wide one-click connect
- nesting / CAM changes

Current Next Task

Implement Milestone 2:

Manual confirmation gate for relationship-based cut

Requirements:

1. Add `verification.level = "manual_confirmed"`.
2. Add helper to convert `bbox_candidate` → `manual_confirmed`.
3. `manual_confirmed` sets:
 - `safeForPreview` = true
 - `safeForCut` = true
 - `requiresManualConfirmation` = false
4. Add Developer UI button:
"Confirm First Valid Relationship For Cut"
or "Confirm Selected Relationship For Cut".
5. Store confirmed relationship in frontend debug state only.

6. Do not persist confirmed relationship to production metadata yet.
7. Preview route continues to accept `bbox_candidate`.
8. Cut validation rejects `bbox_candidate`.
9. Cut validation accepts `manual_confirmed`.
10. Add tests for verification gating.
11. Do not implement face-level detection in this task.
12. Do not implement hinge / lock / runner.
13. Do not modify generators.

Acceptance criteria:

- Offline regression passes.
- Existing relationship tests pass.
- Existing hardware preview tests pass.
- `bbox_candidate` is not cut-safe.
- `manual_confirmed` is cut-safe.
- Developer UI can scan, preview, confirm, and show updated JSON.
- No actual Fusion cut is required in this milestone.